MILITARY SPECIFICATION

VALVE, FUEL SELECTOR

This specification is approved for use by US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification describes a three position rotor type fuel selector valve capable of one to three positions "on" operation with internal detents to positively locate each position.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GSS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 4820

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SPECIFICATIONS
FEDERAL

QQ-P-416	- Plating, Cadmium (Electrodeposited).
TT-S-735	- Standard Test Fluid, Hydrocarbon.
VV-F-800	- Fuel Oil, Diesel.

MILITARY

MIL-H-3136	- Standard Test Fluids, Hydrocarbons.
MIL-G-5572	- Gasoline, Aviation Grades 80/87, 100/130, 115/145.
MIL-T-5624	- Turbine Fuel, Aviation Grade JP4 and JP5.
MIL-P-6871	- Plating, Chromium.
MIL-A-8625	- Anodic Coatings for Aluminum and Aluminum Alloys.

STANDARDS
MILITARY

MIL-STD-105	- Standard Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-202	- Test Method for Electronic and Electronic Components.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-1523	- Age Control of Age-Sensitive Elastrometric Material (for Aerospace Application).
MIL-STD-45662	- Calibration Systems Requirements.
MS 29513	- Packing, Preformed Hydrocarbon Fuel Resistant, O Rings.
MS 33540	- Safety Wiring and Cotter Pinning, General Practices for.

2.1.2 Other Government documents, drawings, and publications. The following documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS
ARMY

8744013	- Valve, Rotor Type Selector, Fuel.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish fuel selector valves which shall be subjected to first article inspection (see 4.4). First article inspection samples, properly marked with identifying information shall be representative of the unit to be furnished to the Government. All subsequent selector valves delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Materials. Material and processes used for the manufacture of fuel selector valves shall be of high quality, suitable for the purpose, and shall conform to applicable Government specifications. The use of contractor specifications shall not constitute waiver of Government specifications. All materials used in the valve shall be sufficiently resistant to MIL-T-5624, MIL-G-5572 and VV-F-800 fuels, with aromatics of 0 to 30 percent to assure satisfactory operation as herein described (see 4.6.1).

3.2.1 Metal parts. Copper and copper alloy parts are not permitted in the manufacture of this valve (see 4.6.1).

3.2.1.1 Protective treatment. Steel parts, other than corrosion resisting steels, not in moving contact shall be cadmium plated in accordance with QQ-P-416, type II. Sliding or rotating steel parts shall be chromium plated in accordance with MIL-P-6871 or adequately treated in some other acceptable manner for corrosion protection where the galling of cadmium plating would cause malfunctioning. Stainless steel parts shall be passivated (see 4.6.1).

3.2.1.2 Anodizing. All aluminum parts shall be anodized in accordance with MIL-A-8625 (see 4.6.1).

3.2.2 Non-metallic parts.

3.2.2.1 Age control. The assembly date of an assembly containing any synthetic age sensitive elastometers as defined by MIL-STD-1523 shall be permanently marked per MIL-STD-1523 requirements and affixed to the component (such that obliteration shall not result from normal use or maintenance) after assembly of the component. All age sensitive elastomers shall be installed in the component within eight quarters after the cure date (see 4.6.1).

3.2.2.2 "O" rings. All "O" rings shall conform to MS 29513 (see 4.6.1).

3.3 Design and construction. The valves shall be designed to permit selection of various fixed fuel-flow patterns or to act as a shut off valve (see 4.6.2).

3.3.1 Configuration. The valve shall conform to Drawing 8744013 (see 4.6.2).

3.3.2 Standard parts. Standard parts shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part number (see 4.6.2).

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3.3.3 Fasteners. All screws shall be interlocked as specified in MS 33540 (see 4.6.2).

3.3.4 Interchangeability. All parts having the same manufacturer's part number shall be completely interchangeable with each other with respect to installation and performance (see 4.6.2).

3.3.5 Lubrication. It is desired that the valve operate satisfactorily without the use of lubricants, for any moving part, other than fuel in the valve. Valves requiring lubrication of any moving part shall be designed and constructed to provide adequate protection to insure that the lubricant will be retained at all times during normal operation and during all of the tests specified herein, and care shall be taken to prevent the use of an excess of lubricant which could adversely affect low temperature operation. The valve shall be capable of operating a reasonable number of cycles dry as specified in the Dry endurance test herein (see 4.6.2).

3.3.6 Direction of rotation. The fuel selector valve shall operate satisfactorily when turned in either a clockwise or counterclockwise direction of rotation (see 4.6.2).

3.3.7 Ports. All ports shall be integral with the valve body (see 4.6.2).

3.3.8 Port positioning. Means shall be provided as a part of the valve for assuring that, with the drive shaft set at a given index position, the desired port or ports will be fully open. There shall be a definite "feel" when the valve enters, leaves, or goes through an index position. There shall be no false index feel between any two index positions. The relation between the body and the movable parts shall not change during normal operation when at a given index position. It is desirable that indexing mechanism be readily removable without disturbing the hydraulic parts of the valve (see 4.6.3).

3.4 Performance.

3.4.1 Selector valve torque. The operating starting torque shall not exceed 25 inch-pounds (in-lb). The running torque shall not exceed 15 in-lb (see 4.6.4).

3.4.2 Operating pressure. The valve shall operate at 30 psi, and withstand a proof pressure of 60 pounds per square inch (psi) with a minimum burst pressure of 90 psi (see 4.6.5).

3.4.3 Flow rate. The rated flow of the valve with three ports open shall exceed 3.7 gallons per minute (gpm), 220 gallons per hour (gph), with a pressure drop not to exceed 0.5 psi (see 4.6.6).

3.4.4 Leak rate. Internal leakage shall not exceed 10 drops per minute at 30 psi (see 4.6.7).

3.4.4.1 External leakage. There shall be no external leakage at pressure up to 60 psi (see 4.6.5.2).

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3.4.5 Hydraulic lock. A hydraulic lock shall not develop under any conditions of operation including plugged ports (see 4.6.8).

3.4.6 Endurance. Endurance shall exceed 10 000 cycles representing approximately 5000 revolutions of the valve in each direction (see 4.6.9).

3.5 Environmental requirements.

3.5.1 Corrosion resistance. Selector valve performance shall not be adversely affected by corrosion when exposed to a 5 percent salt spray for 200 hours (see 4.6.10).

3.5.2 Humidity. The selector valve shall meet the requirements of 3.4.3 after storage under induced high temperature and humidity conditions (see 4.6.11).

3.5.3 Vibration. After being tested as specified in 4.6.12, the valve shall evidence no permanently deformed, cracked, or broken parts and shall meet the requirements of 3.4.3.

3.5.4 Fungus. The valve shall meet the requirements after exposure to conditions favorable to fungus growth (see 4.6.13).

3.5.5 Extreme temperature resistance.

3.5.5.1 High temperature. The valve shall meet the requirements of 3.4.3 after 2000 revolutions at 125 to 135 degrees Fahrenheit (°F) (see 4.6.14).

3.5.5.2 Low temperature. After 72 hours storage at -65 to -70°F, the pump shall meet the requirements of 3.4.3 after 500 valve revolutions (see 4.6.15).

3.6 Identification and marking. Identification and markings shall meet the requirements of MIL-STD-130 (see 4.6.2).

3.7 Workmanship. Workmanship shall be such as to provide a product free of defects which would affect its performance or serviceability (see 4.6.2).

3.8 Storage. The valve shall be functional after 3 years storage when packaged according to MIL-P-116 (see 5.1 and 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the

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Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10 percent of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.2 Classification of inspection:

a. First article inspection (see 4.4).
b. Quality conformance inspections (see 4.5).
   1. Examination (see 4.5.2).
   2. Acceptance tests (see 4.5.3.1).
   3. Control tests (see 4.5.3.2).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature 23 $\pm$ 10°C
b. Barometric pressure 725 + 50mm Hg
   - 75
c. Relative humidity 50 $\pm$ 30 percent

4.4 First article inspection. Unless otherwise specified (see 6.2), the Government shall select six fuel selector valves produced under the production contract for first article inspection. First article samples shall be inspected as specified in table I. Approval of the first article sample by the Government shall not relieve the contractor of his obligation to supply fuel selector valves that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

4.4.1 First article inspection failure. Test item deficiencies during, or as a result of, the first article test, shall be cause for rejection of the items until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of the first article test, shall be prima facie evidence that all items already produced prior to completion of the first article test are similarly deficient unless evidence satisfactory to the contracting officer is furnished by the contractor that they are not similarly deficient. Such deficiencies on all items shall be corrected by the contractor at no cost to the Government. The Government shall not final accept products until first article testing is completed to the satisfaction of the Government.

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TABLE I. Classification of inspections.

Title	Requirement	Inspection	First article	Quality conformance		
				Examination	Acceptance (100%)	Control
Material and construction	3.2 thru 3.2.2.2	4.6.1	X			X
Defects (see table II)	3.3 thru 3.3.7, 3.6, and 3.7	4.6.2	X	X		X
Features	3.3.8	4.6.3	X			
Valve torque	3.4.1	4.6.4	X		X	X
Operating pressure	3.4.2	4.6.5	X			X
Flow rate	3.4.3	4.6.6	X		X	X
Leak rate	3.4.4	4.6.7	X			X
External leakage	3.4.4.1	4.6.5.2	X			X
Hydraulic lock	3.4.5	4.6.8	X			
Endurance	3.4.6	4.6.9	X			
Corrosion	3.5.1	4.6.10	X			
Humidity	3.5.2	4.6.11	X			
Vibration	3.5.3	4.6.12	X			
Fungus	3.5.4	4.6.13	X			
High temperature	3.5.5.1	4.6.14.1	X			
Low temperature	3.5.5.2	4.6.15	X			

4.5 Quality conformance inspection.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all the selector valve assemblies of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105.

4.5.2 Quality conformance examinations.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following acceptable quality levels (AQL) on the basis of percent defective (see 6.4).

Classification	AQL
Major	1.0
Minor	2.5

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4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table II.

TABLE II. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 1.0% Defective	
101	Incomplete, assembly (see 3.3.1).	Visual
102	Dimensions affecting interchangeability, out of tolerance (see 3.3.1 thru 3.3.4).	SIE 1/
103	Identification marking, improper (see 3.6).	Visual
104	Faulty workmanship affecting performance (see 3.7).	Visual
Minor	AQL 2.5% Defective	
201	Incorrect dimensions not affecting interchangeability, out of tolerance (see 3.3.1).	SIE 1/
202	Faulty workmanship not affecting interchangeability (see 3.7).	Visual

1/ SIE = Standard Inspection Equipment.

4.5.3 Quality conformance tests.

4.5.3.1 Acceptance tests (100 percent). Each valve assembly shall be subjected to the acceptance tests specified in table I.

4.5.3.2 Control tests. Control tests shall be conducted on one valve assembly from each lot of 200 units consecutively produced, except that not more than one test shall be performed in a three month period, nor less than one test in a six month period. The valve assemblies shall be subjected to the control tests specified in table I.

4.5.3.3 Failure. Failure of any valve assembly to pass any of the specified inspections shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.6 Methods of inspection.

4.6.1 Materials and construction. Conformance to 3.2 through 3.2.2.2 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

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4.6.2 Defects. Conformance to 3.3 through 3.3.7 shall be determined by examination for the defects listed in table II. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

4.6.3 Features. Conformance to 3.3.8 shall be determined by qualitative observation sufficient to demonstrate that specified characteristics and features are functional.

4.6.4 Operating valve torque. To assure conformance to 3.4.1, the valve shall be wetted with fuel, and the torque required to start the valve from all indexing positions and between all index positions in both directions of rotation shall be measured. The starting torque shall not exceed 25 lb-in and the running torque shall not exceed 15 lb-in.

4.6.5 Operating pressure. To determine conformance to 3.4.2, the valve shall be tested with three open ports producing a fluid flow rate in excess of 3.7 gpm. The operating pressure shall be 30 psi with the pressure drop not to exceed 0.5 psi.

4.6.5.1 Burst pressure. With the valve in the closed position a fluid pressure of 90 psi shall be applied successively for one minute to each port. There shall be no evidence of damage or distortion (see 3.4.2).

4.6.5.2 Proof pressure. Following the burst proof of 4.6.5.1 the test pressure shall be lowered to 60 psi, there shall be no evidence of external leakage of the valve (see 3.4.2 and 3.4.4).

4.6.6 Flow rate. To determine conformance to 3.4.3, the rated flow shall be determined in the three port open position at a fluid pressure of 30 psi. The flow rate shall exceed 3.7 gpm, 220 gph with a measured pressure drop not to exceed 0.5 psi.

4.6.7 Internal leakage. Each valve port shall be tested separately with fuel under pressure of 30 psi. The tests shall be conducted with all open ports plugged, except the port to which pressure is being applied. If leakage is detected it shall not exceed 10 drops (5 cc) per minute at 30 psi for total leakage of all ports of the valve (see 3.4.4).

4.6.8 Hydraulic lock. To determine conformance to 3.4.5 verify that a hydraulic lock does not develop during the tests specified in 4.6.5.

4.6.9 Endurance. To conform to the 10 000 cycle endurance requirement of 3.4.6 the valve shall be turned at an average speed of 8 revolutions per minute (rpm) with a minimum instantaneous speed of 30 rpm. The valve may be turned by hand using a suitable torque wrench or by a suitable mechanical drive that will automatically shut off the drive upon overload. The direction of rotation shall be reversed periodically so that the valve is rotated an approximate equal number of times in each direction.

4.6.10 Corrosion resistance test. To determine conformance to 3.5.1, the selector valve shall be subjected to method 10; 20 percent salt spray of MIL-STD-202 for a period of 100 hours.

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4.6.11 Humidity tests. To determine conformance to 3.5.2, the valve shall be tested in accordance with MIL-STD-810, method 507.2, procedure II. Storage ambient conditions shall conform to cycle 4, figure 507.2.2.

4.6.12 Vibration. To determine conformance to 3.5.3, the item shall be tested in accordance with MIL-STD-810, method 514.3, procedure I. It shall be solid mounted to the vibration fixture only by means of the four bolt holes of the mounting flange. A 20 minute logaritmic sweep shall be conducted in each of three mutually perpendicular asxes, from 5 to 500 hertz (Hz), to identify resonant frequencies. A resonant dwell shall be conducted at each resonant frequency for 20 minutes in each axis (see figure 1).

4.6.13 Fungus. To determine conformance to 3.5.4, the item shall be tested in accordance with MIL-STD-810, method 508.3. In lieu of a test, certification, with supporting data, may be provided attesting that the item is constructed of materials which will not support fungus growth.

4.6.14 High temperature resistance. The valve shall be mounted in a suitable test fixture with one fluid flow path open. The valve shall be soaked in TT-S-735, type III at 130 ± 5°F for a period of 72 hours. Immediately after the soak and while still at the specified temperature the valve shall be tested for the leakage of the valve body, and shaft seal, and for interport fuel pressure and air suction and normal torque (see 3.5.5.1).

4.6.15 Low temperature resistance. Using fuel conforming to MIL-H-3136, type I, the valve shall be subjected to the tests specified in 4.6.14 except the storage temperature shall be -65 to -70°F. The pump shall meet the requirements of 3.4.3 after 500 valve revolutions at ambient temperature (see 3.5.5.2).

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging standard or packaging data sheet specified by the contracting authority (see 6.2).

6. NOTES

6.1 Intended use. These fuel selector valves are intended for use in tank-automotive fuel systems. They are intended to be manually operated. When operated by other means, additional tests and separate approval may be required by the procuring activity.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type, class [drawing no.] of required item (see 1.2).
c. If first article samples are not required (see 3.1).
d. If responsibility for inspection shall be other than as specified (see 4.1).
e. If responsibility for inspection equipment shall be other than as specified (see 4.1.1).

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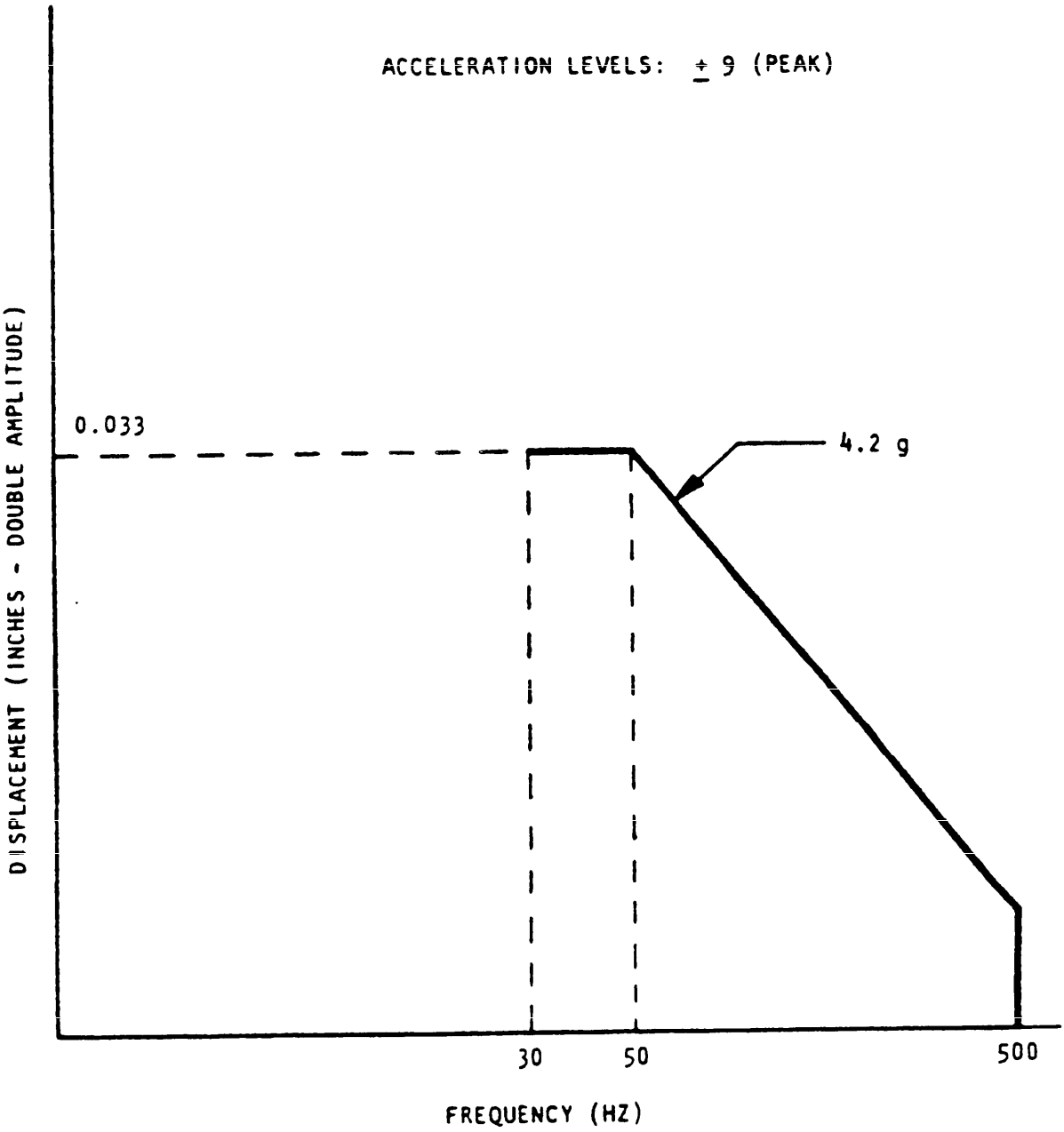
f. If inspection conditions shall be other than as specified (see 4.3).
g. Selection of applicable level of preservation, packaging, packing, and marking (see 5.1).

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NOTE: All curves shall be extended to 2 Hz. when test item resonances below 5 Hz. are expected.

FIGURE 1. Vibration test curve.

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